Unleashing UVM™

STACK MEMORY SPECIFICATION

SystemVerilog, UVM - training, methodology consulting, auditing & debug

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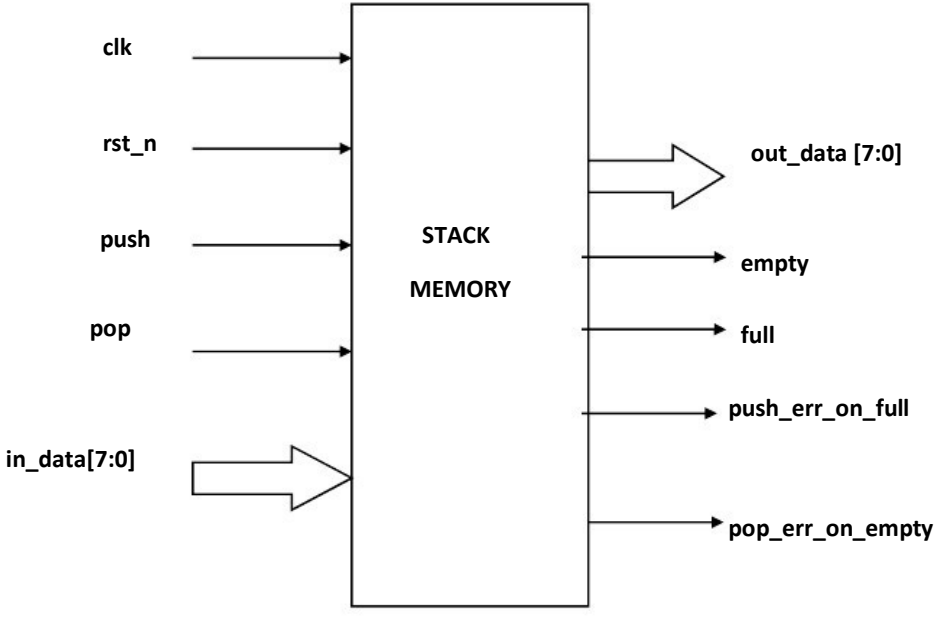
STACK MEMORY :



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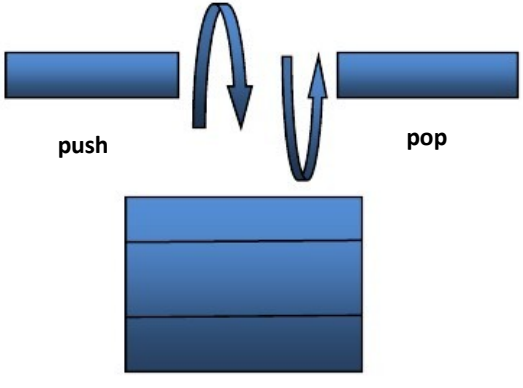
STACK VIEW :



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Signal List:

SIGNAL	DIRECTION	WIDTH	DESCRIPTION
clk	Input	1 bit	Clock of 50 MHz [assumption].
rst_n	Input	1 bit	Active_low_reset which Reset the design.
in_data	Input	8 bits	Input data into the stack.
push	Input	1 bit	Enables push data to the stack.
pop	Input	1 bit	Enables pop data from the stack.
out_data	output	8 bits	output data from the stack.
full	output	1 bit	Indicates that the stack is full.
empty	output	1bit	Indicates that the stack is empty.
push_err_on_full	output	1bit	Indicates error if trying to push data when stack is full
pop_err_on_empty	output	1 bit	Indicates error if trying to pop data when stack is empty

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CONTROL PATH DESIGN

STACK works as LIFO (Last in First out)

• PUSH & POP OPERATIONS

• PUSH
– Push signal causes PUSH Operation.
– Data into STACK.

• POP
– Pop signal causes POP Operation.
– Data extracted from STACK.

STACK depth=1K

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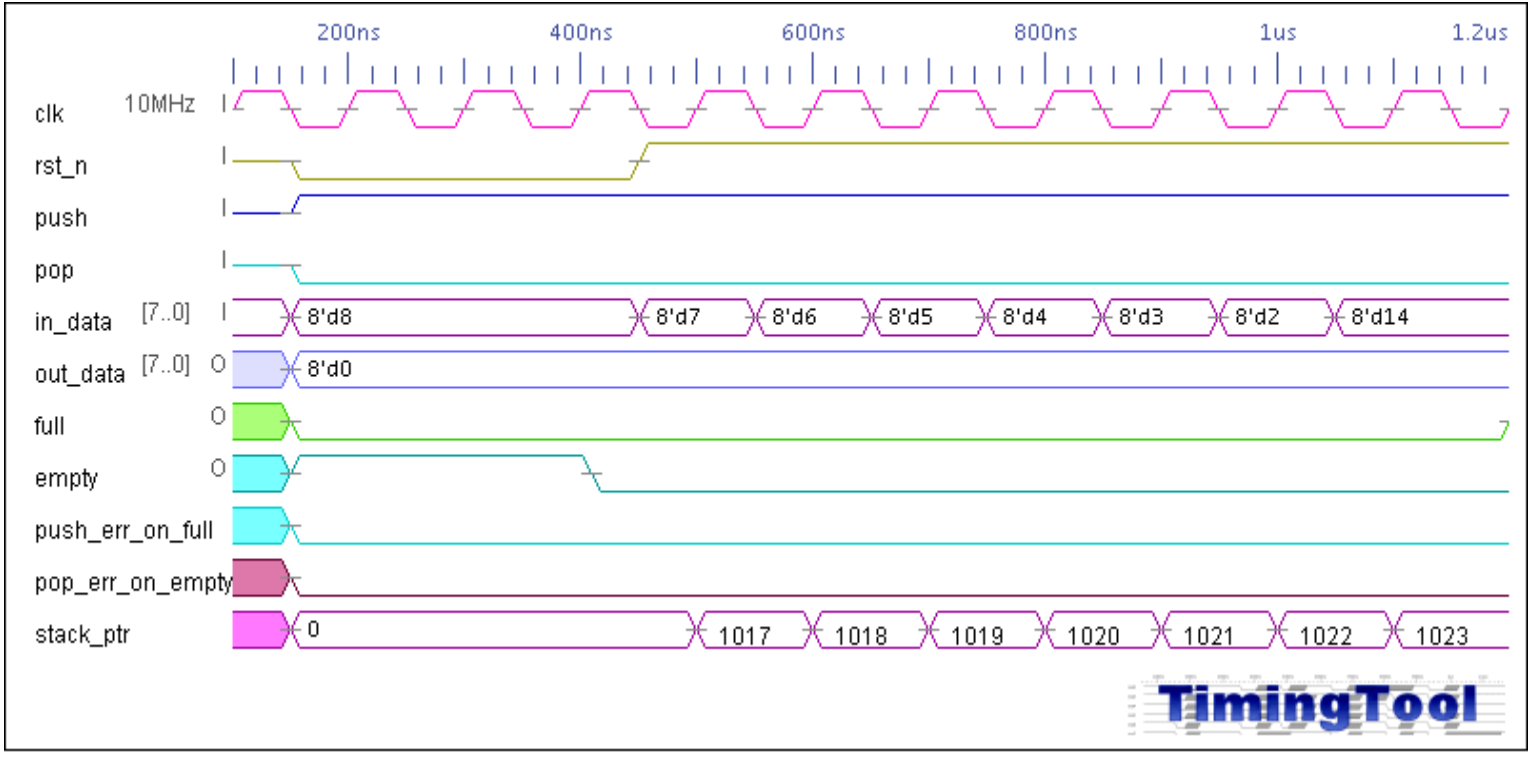
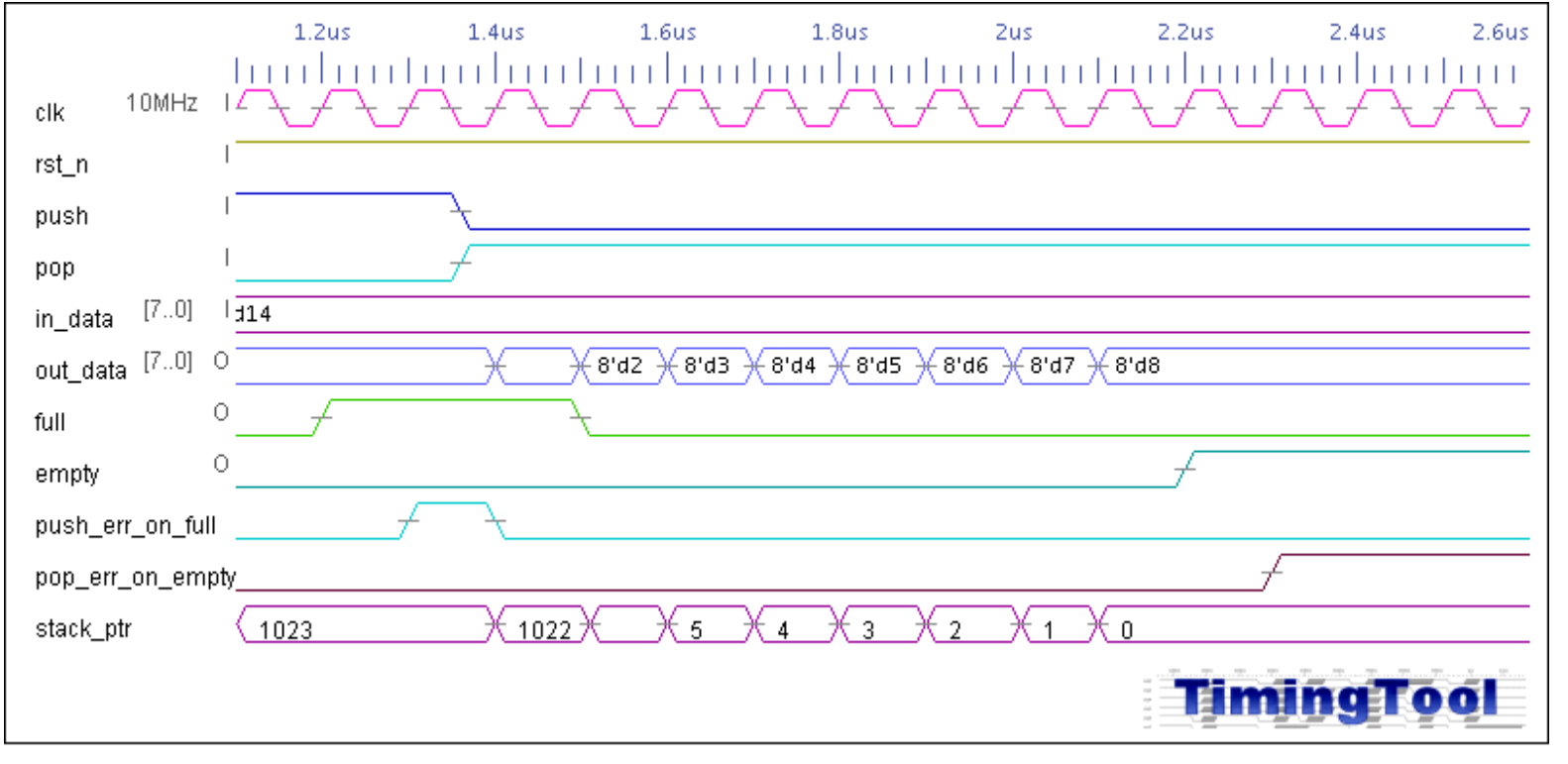
Timing_Diagram :



Timing_Diagram :